GenC Program

IDE Java FSE GenC Next - Angular - Handbook

Program Overview

The **IDE Java FSE GenC Next - Angular training** is a 12.08-week immersive program designed to build full-stack engineering capabilities through a blend of:

- Self-learning modules
- Hands-on exercises
- Agile ceremonies
- Classroom sessions (Technical & Behavioural)
- Capstone project - Integrated Development Project (IDP)

All learning materials and exercises are hosted on the [GenC Learn] platform.

What is DevLed Skilling?

The **DevLed Skilling Program** is a transformative, project-based training initiative designed to simulate a real-world software development environment. Unlike traditional training models, DevLed immerses participants in an agile, collaborative, and delivery-focused setting from day one-mirroring the exact conditions of live customer projects.

Why DevLed?

- **Real-World Simulation:** Participants work in agile teams, attend ceremonies, and follow sprint cycles just like in actual delivery projects.
- **End-to-End Exposure:** From UI development to backend services, testing, logging, and DevOps, trainees experience the full software lifecycle.
- **Customer-Centric Mindset:** The training emphasizes delivering value to stakeholders, fostering a strong sense of ownership and accountability.
- **Hands-On Learning:** With guided exercises, classroom sessions, and a capstone project, learning is deeply practical, and outcome driven.
- **Day-One Billability:** By the end of the program, participants are equipped with the skills, mindset, and experience to contribute to real projects immediately ensuring day-one billability.

DevLed Training Structure (Agile Way)

Sprint	Duration	Phase Name	Modules/Courses Covered
Sprint 0	0.6 week	Project Environment Setup	• Agile Methodology • Jira Basics • GIT Basics
Sprint 1	1.31 weeks	Backend Foundations	• Design Patterns and Principles • Reactive Programming with Java
Sprint 2	3 weeks	API Development and Testing	• Reactive Programming Using Spring Webflux Framework • TDD using JUnit and Mockito • SLF4J and Lombok • Spring Data JPA with Spring Boot, Hibernate • Spring REST using Spring Boot 3
Sprint 3	2 weeks	Microservices & Containerization	• Microservices with Spring Boot 3 and Spring Cloud • Docker Basics
Sprint 4	4 weeks	Front-end Development, DevOps and Cloud, Code Quality	• Node.js • TypeScript • Bootstrap 5 • Angular (v20.0) • Application Debugging - Front-end • Code Quality Standards • SonarQube • Cloud and DevOps Basics using AWS

Learning Components

1. Self-Paced Learning Materials (Udemy, Open-Source Learning Links)

- Hosted on **GenC Learn Learning Path – ADM-DE Java FSE - Angular (GenC Next)**
- Includes videos, readings, and quizzes
- **Note: It is NOT mandatory to complete the entire Udemy courses**. Learners are encouraged to **refer to specific sections that align with the training scope** for reinforcement and continuous learning.

Recommended Udemy Learning

The following section outlines the recommended external learning resources to supplement the core curriculum. While completing the entire course is not mandatory, learners are highly encouraged to utilize these as a secondary reference.

To optimize study time, GenCs should focus on the specific modules or sections of the Udemy courses that align directly with the technical scope of the current curriculum. These resources serve as a valuable tool for reinforcing conceptual understanding and providing additional practical examples beyond the classroom sessions.

Udemy Learning Mapping

Sprint	Course/Module	Udemy Course Name
Sprint 0	Agile Methodology	Agile Crash Course: Agile Project Management; Agile Delivery
	Jira Basics	Jira for Beginners - Detailed Course to Get Started in Jira
	GIT Basics	Git Complete: The definitive, step-by-step guide to Git
Sprint 1	Reactive Programming with Java	Mastering Java Reactive Programming [From Scratch] Reactive Programming in Modern Java using Project Reactor
	Design Patterns and Principles	Complete Java Design Patterns masterclass SOLID Principles: Introducing Software Architecture & Design
Sprint 2	Reactive Programming Using Spring Webflux Framework	Reactive Programming with Spring Framework 5 Reactive Applications with Spring WebFlux Framework
	TDD using JUnit and Mockito	Learn Java Unit Testing with Junit & Mockito in 30 Steps Spring Boot Unit Testing with JUnit, Mockito and MockMvc
	SLF4J and Lombok	Java Best Practices for Efficient, Scalable, and Secure Code

	Spring Data JPA with Spring Boot, Hibernate	Hibernate and Spring Data JPA: Beginner to Guru
	Spring REST using Spring Boot 3	Building Real-Time REST APIs with Spring Boot - Blog App
	Application Debugging - Backend	Eclipse Debugging Techniques And Tricks
Sprint 3	Microservices with Spring Boot 3 and Spring Cloud	[NEW] Building Microservices with Spring Boot & Spring Cloud
	Docker Basics	Docker for the Absolute Beginner - Hands On - DevOps
Sprint 4	Node.js Fundamentals	The Complete Node.js Developer Course (3rd Edition)
	TypeScript Basics	Understanding TypeScript
	Bootstrap 5	The Ultimate Bootstrap Guide - Bootstrap 5 from Scratch
	Angular (v20.0)	Angular - The Complete Guide Angular Testing Masterclass (Angular 20)
	Application Debugging - Front-end	Devtools Pro: The Basics of Chrome Developer Tools
	Cloud and DevOps using AWS	Introduction to Cloud Computing on AWS for Beginners [2025] DevOps, CI/CD(Continuous Integration/Delivery) for Beginners AWS CodePipeline Step by Step
	Code Quality Standards	Java Best Practices for Efficient, Scalable, and Secure Code
	SonarQube	Continuous Code Inspection with SonarQube

2. Hands-On Exercises

- Available in the **GenC Learn Learning Path – ADM-DE Java FSE - Angular (GenC Next)**
- **Mandatory** for each sprint
- Includes key programming language and framework related aspects, coding challenges, and Integrated Capability Test
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3. Classroom Sessions

- Daily sessions conducted by technical **SMEs**
- Focus on concept reinforcement, Q&A, and live coding
- **Attendance** is mandatory

4. Capstone Project

- Begins in Sprint 1 and continues through Sprint 4
- **Team-based** implementation of a full-stack application
- Evaluated on design, code quality and testing

Agile Ceremonies

Participants are expected to engage in:

- Daily Stand-ups (15 minutes)
- Sprint Planning (1 hour)
- Sprint Reviews and Retrospectives (1 hour)

These ceremonies simulate real-world agile environments and are essential for collaboration and progress tracking.

Evaluation Framework

To assess progress and readiness, the training includes **two formal evaluations** conducted by **BU Subject Matter Experts (SMEs)** on a one-on-one basis. These evaluations cover both technical competencies and project contributions.

1. Interim Evaluation

- **Timing:** After Sprint 3 (around Week 8)
- **Focus Areas:**
    - **Technical:** Foundational skills, agile practices, backend, data access and testing
    - **Project:** Hands-on completion, collaboration, initial project setup
- **Format:** One-on-one technical and project review
- **Outcome:** RAG (Red-Amber-Green) grading
    - **Green:** On track and progressing well
    - **Amber:** Needs improvement in specific areas
    - **Red:** Significant gaps in understanding or delivery

2. Final Evaluation

- **Timing:** End of Sprint 4 (Week 12)
- **Focus Areas:**
    - **Technical:** Full-stack development, microservices, code quality, DevOps
    - **Project:** Capstone project delivery, architecture, implementation, testing, deployment (local)
- **Format:** One-on-one technical interview and project walkthrough

- **Outcome:** RAG grading
    - Green status is expected for successful completion

Expectations & Completion Criteria

- Learn all self-learning modules
- Submit all mandatory hands-on exercises
- Participate in all agile ceremonies
- Attend classroom sessions
- Clear Interim and Final evaluations
- Successfully deliver the capstone project
- Clear **Final HackerRank** assessment (After successful completion of final evaluation)